DATA SCIENCE ENIGMA: THE ARCANE BLAST

Background Scenario

You have just been hired as a Data Scientist for a massive multiplayer online role-playing game (MMORPG). The game designers recently introduced a highly anticipated wizard spell called *Arcane Blast*. However, the player base is in uproar, complaining that the damage output is wildly inconsistent—sometimes it instantly defeats enemies, and other times it barely scratches them.

The lead game designer insists the spell is not broken. They claim the damage is calculated using a strict, unvarying mathematical formula based on the wizard's stats, the monster's defense, and the exact spatial grid positions of both entities. There is absolutely zero randomness (RNG) in the calculation.

They have handed you a dataset of 5,000 recent spell casts (`arcane_blast_logs.csv`) and issued a challenge: **Reverse-engineer the exact mathematical formula.**

The Dataset

The dataset contains the following variables recorded for each spell cast:

Variable Name	Description	Typical Range
`Player_X`	The X-coordinate of the wizard on the game grid	0 to 100
`Player_Y`	The Y-coordinate of the wizard on the game grid	0 to 100
`Enemy_X`	The X-coordinate of the monster on the game grid	0 to 100
`Enemy_Y`	The Y-coordinate of the monster on the game grid	0 to 100
`Base_Spell_Power`	The wizard's core magic stat	50 to 200
`Enemy_Armor`	The monster's physical defense stat	10 to 50
`Damage_Dealt`	**Final damage taken by the monster (Target)**	**0 to ∼10,000**

The Challenge Constraints

To prove your mastery over foundational Machine Learning concepts, you must solve this using the most basic tool available, adhering strictly to the following rules:

- **Algorithm Restriction:** You are **ONLY** allowed to use standard Ordinary Least Squares (OLS) Linear Regression (e.g., `sklearn.linear_model.LinearRegression`). You may not use Decision Trees, Random Forests, Support Vector Machines, Neural Networks, or any gradient boosting models.
- **Accuracy Threshold:** Your final model must achieve an R^2 score of **at least 0.98** on a hidden test set.

A Parting Hint

If you pass the raw coordinate data directly into a Linear Regression model, it will fail catastrophically. The algorithm will treat the variables as independent, parallel lines. But you are not dealing with parallel lines.

> *"Coordinates themselves don't hurt people. Geometry hurts people."*
> — **The Lead Designer**

Can you manipulate the spatial numbers, strictly using your knowledge of mathematics, algebra, and geometry, to crack the code and balance the game?